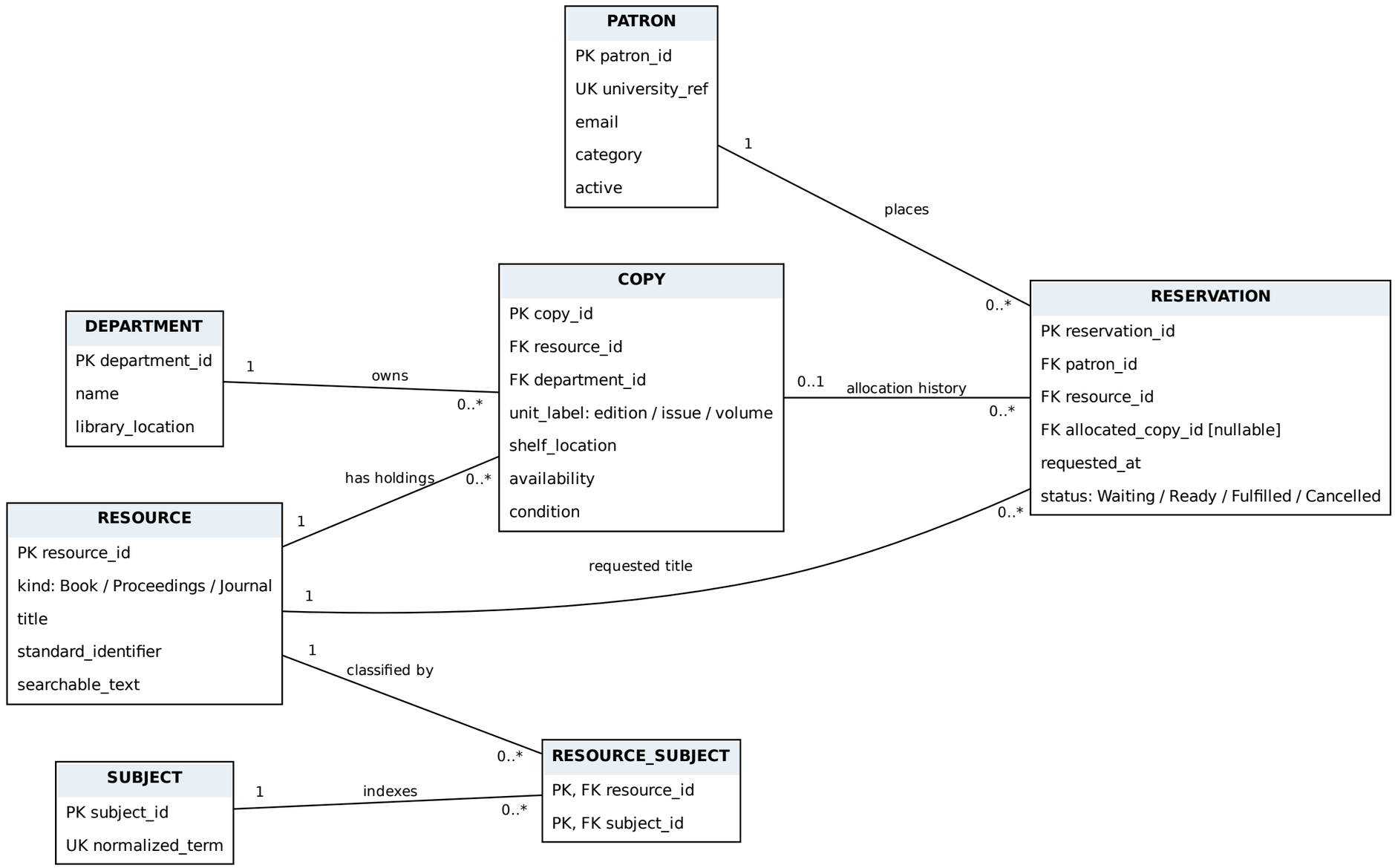
# S1 — Entity Relationship Diagram

Search the unified catalogue and reserve a resource

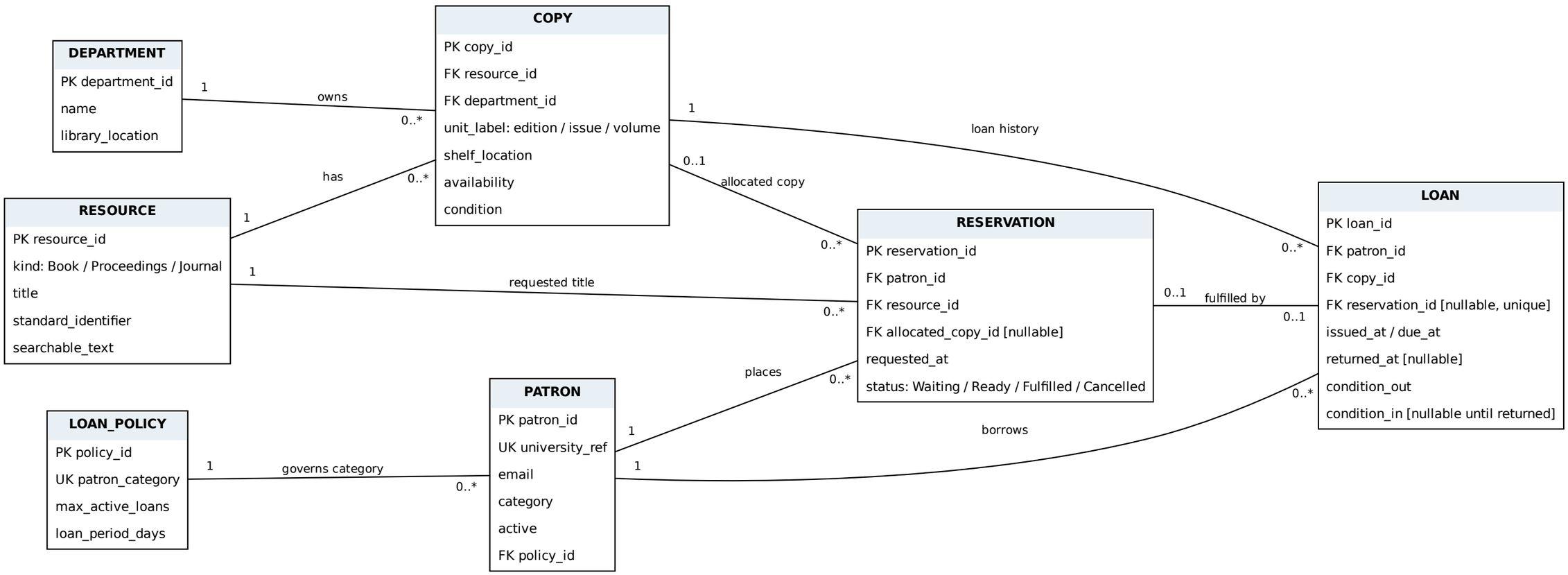



PK = primary key; FK = foreign key; UK = unique key. Numbers at an entity end show its allowed multiplicity for one entity at the opposite end.

Each Reservation belongs to exactly one Patron and Resource; allocated_copy_id is null while Waiting and required while Ready. Any allocated copy must belong to the same Resource. One copy has at most one active Ready allocation; only an eligible Available copy can become Reserved. RESOURCE_SUBJECT has a composite primary key; subjects are shared across departments.

# S2 — Entity Relationship Diagram

Borrow a reserved copy, return it and preserve borrower history

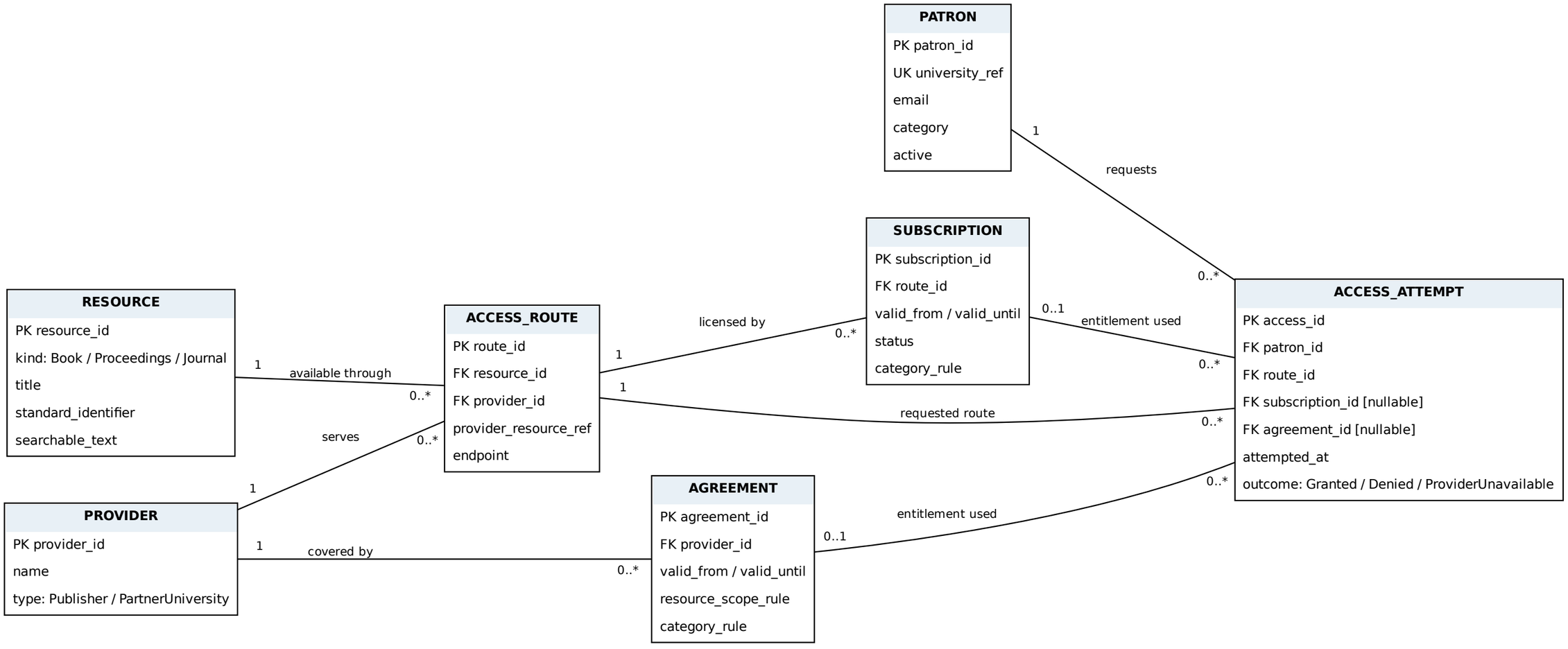



PK = primary key; FK = foreign key; UK = unique key. Numbers at an entity end show its allowed multiplicity for one entity at the opposite end.

In this view PATRON also has FK policy_id -> LOAN_POLICY. A Loan always links one Patron to one Copy; an optional reservation_id is unique and, when supplied, must match the same patron and allocated copy. At most one Loan per Copy can have returned_at null. Keep closed loans and condition records; return does not delete history. Copy becomes Available only if inspection permits circulation.

# S3 — Entity Relationship Diagram

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PK = primary key; FK = foreign key; UK = unique key. Numbers at an entity end show its allowed multiplicity for one entity at the opposite end.

For Granted attempts, exactly one of subscription_id or agreement_id is present (XOR); its route/provider, scope, category rule and validity interval must match the requested access. A Denied attempt may have neither reference. Partner access is represented by PROVIDER.type = PartnerUniversity plus an applicable Agreement. A grant depends on provider acceptance; audit stores outcome and references, not reusable grant tokens.